12 JAN
24
Friday

Assignment-I

1. Let

(a) $\mathcal{C}_1 = \{[a,b] : a,b \in \mathbb{R}\}$ a,b finite

(b) $\mathcal{C}_2 = \{(-\infty, a) : a \in \mathbb{R}\}$

$\mathcal{C}_3 = \{(-\infty, a] : a \in \mathbb{R}\}$

$\mathcal{C}_4 = \{(a, \infty) : a \in \mathbb{R}\}$

$\mathcal{C}_5 = \{[a, \infty) : a \in \mathbb{R}\}$

Now:

~~$(-\infty, b) = \bigcup [n, b + \frac{1}{n}]$~~

- $(-\infty, b) = \bigcup_{n=1}^{\infty} [-n, b - \frac{1}{n}]$

$\Rightarrow (-\infty, b) \in \sigma(\mathcal{C}_1) \Rightarrow \sigma(\mathcal{C}_2) \subseteq \sigma(\mathcal{C}_1)$ — (1)

Now

- $(-\infty, a] = \bigcap_{n=1}^{\infty} (-\infty, a + \frac{1}{n}]$

$\Rightarrow \mathcal{C}_3 \subseteq \sigma(\mathcal{C}_2) \Rightarrow \sigma(\mathcal{C}_3) \subseteq \sigma(\mathcal{C}_2)$ — (2)

- $(a, \infty) = (-\infty, a]^c$

$\Rightarrow$ ~~$\mathcal{C}_4 \subseteq \mathcal{C}$~~ $\mathcal{C}_4 \subseteq \sigma(\mathcal{C}_3)$

$\Rightarrow \sigma(\mathcal{C}_4) \subseteq \sigma(\mathcal{C}_3)$ — (3)

- $[a, \infty) = \bigcup_{n=1}^{\infty} (a - \frac{1}{n}, \infty)$

$\Rightarrow \mathcal{C}_5 \subseteq \sigma(\mathcal{C}_4) \Rightarrow \sigma(\mathcal{C}_5) \subseteq \sigma(\mathcal{C}_4)$ — (4)

- Now, since

$$[b,\infty)^c = (-\infty, b)$$

$$= \bigcup_{n=1}^{\infty} (-\infty, b-\frac{1}{n})$$

$$(-\infty, b] = \bigcup_{n=1}^{\infty} (-\infty, b-\frac{1}{n})$$

$$= \bigcup_{n=1}^{\infty} \left[b-\frac{1}{n}, \infty\right)^c$$

And

$$[a,b] = [a,\infty) \cap \left\{ \bigcup_{n=1}^{\infty} \left[b-\frac{1}{n}, \infty\right)^c \right\}$$

$\Rightarrow [a,b] \in \sigma(\mathcal{C}_5)$

$\Rightarrow \mathcal{C}_1 \subseteq \sigma(\mathcal{C}_5) \Rightarrow \sigma(\mathcal{C}_1) \subseteq \sigma(\mathcal{C}_5)$ — (5)

Now; from (1), (2), (3), (4) & (5)

$$say(S) = \sigma(\mathcal{C}_1) = \sigma(\mathcal{C}_2) = \sigma(\mathcal{C}_3) = \sigma(\mathcal{C}_4) = \sigma(\mathcal{C}_5)$$

Now; $[a,\infty) \in S \quad \forall\ a \in \mathbb{R}$

$(-\infty, b] \in S \quad \forall\ a \in \mathbb{R}$

$\Rightarrow [a,\infty) \cap (-\infty, b] \in S$

$\Rightarrow [a,b] \in S \quad \forall\ a, b \in \mathbb{R}$

$\Rightarrow$ ~~σ(all~~

Sall class set of $\mathbb{R}$S $\subseteq S$

$\Rightarrow \mathcal{B}_{\mathbb{R}} \subseteq S = \sigma(\sigma(\mathcal{C}_1))$

& also, $\mathcal{C}_1 \subseteq \mathcal{B}_{\mathbb{R}} \Rightarrow \sigma(\mathcal{C}_1) \subseteq \mathcal{B}_{\mathbb{R}}$

$\Rightarrow \sigma(\mathcal{C}_1) = \mathcal{B}_{\mathbb{R}}$

i.e.

$$\sigma(\mathcal{C}_1) = \sigma(\mathcal{C}_2) = \sigma(\mathcal{C}_3) = \sigma(\mathcal{C}_4) = \sigma(\mathcal{C}_5) = \mathcal{B}_{\mathbb{R}}$$

13-J Apr 2023

Q.2. Prove that

$\mathcal{C}_e = \{$ finite disjoint union of collection $(a, b]$, $a, b \in \mathbb{R}\}$

Now:

a) Since $\mathbb{R} = (-\infty, \infty]$

$= (-\infty, \infty)$

$\Rightarrow \mathbb{R} = \Omega \in \mathcal{C}_e$

b) Let $A \in \mathcal{C}_e$

where $A = B_1 \cup B_2 \cup B_3 \cup \dots \cup B_n$

Now: say $B_1 = (a_1, b_1]$

$B_2 = (a_2, b_2]$

Now,

$B_1 \cup B_2 = (a_1, b_1] \cup (a_2, b_2]$ where $B_1 \cap B_2 = \phi$ when $b_1 < a_2$

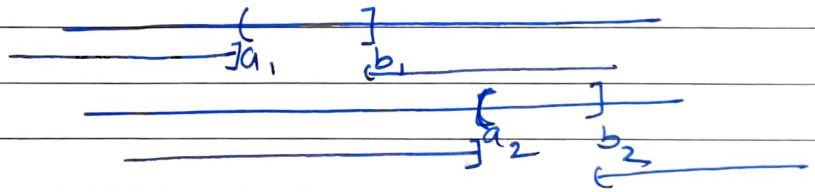
Now:

$B_1^c = (-\infty, a_1] \cup (b_1, \infty)$

$B_2^c = (-\infty, a_2] \cup (b_2, \infty)$

Now:

$B_1^c \cap B_2^c = (-\infty, a_1] \cup (b_1, a_2] \cup (b_2, \infty)$



Now, B_1^c similarly, we see by Induction

$B_1^c \cap B_2^c \cap \dots \cap B_n^c \in \mathcal{C}_e$

$\Rightarrow (B_1 \cup B_2 \dots \cup B_n)^c \in \mathcal{C}_e$

$\Rightarrow A^c \in \mathcal{C}_e$.

c) Let $A_1, A_2, \dots A_n \in \mathcal{C}_e$

each A_i is finite disjoint union of B_i's

So $\bigcup_{j=1}^{n} A_j = \bigcup_{j=1}^{n} \left(\bigcup_{i=1}^{m_i} B_i^j \right)$: where $A_j = \bigcup_{i=1}^{m_i} B_i^j$

$\Rightarrow \mathcal{C}_e$ is a field.

Q.3. Let (Ω, f) be a measure space

- for $x \in \mathbb{R}$,
  $\delta_x : f \longrightarrow [0, \infty]$
  s.t.
  $$\delta_x(A) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

Now,
Let $A_1, A_2, \ldots A_n \ldots \in f$ be disjoint collection of subsets
Now: Let $x \in \bigcup_{i=1}^{\infty} A_i$

$\Rightarrow x \in A_j$ for exactly one j.
$\Rightarrow \delta_x(A_j) = 1$ & $\delta_x(A_j) = 0 \quad \forall\ i = 1, 2, \ldots \neq j$
ab
$$\delta_x\left(\bigcup_{i=1}^{\infty} A_j\right) = 1$$

$$\Rightarrow \delta_x\left(\bigcup_{i=1}^{\infty} A_i\right) = 0 + 0 + \ldots + 1 + 0 + \ldots$$
$$= \delta_x(A_1) + \ldots + \delta_x(A_j) + \ldots$$
$$= \sum_{i=1}^{\infty} \delta_x(A_i)$$

if $x \notin \bigcup_{i=1}^{\infty} A_i \Rightarrow x \notin A_i \quad \forall\ i = 1, 2, \ldots n$

$$\Rightarrow \delta_x\left(\bigcup_{i=1}^{\infty} A_i\right) = 0 = \sum_{i=1}^{\infty} \delta_x(A_i) \quad \left\{ \because \delta_x(A_i) = 0 \text{ for } i = 1, 2, \ldots n \right.$$

- Define $\mu : f \longrightarrow [0, \infty]$
  s.t. $\mu(A)$ = Number of elements in A.

Let
$A_1, A_2, \ldots A_n \ldots \in f$ be disjoint subsets of Ω. s.t. Ω,

Now, let $|A_i| = \eta_i$

$\Rightarrow$ Now we see that

$$\left|\bigcup_{i=1}^{\infty} A_i\right| = \sum_{i=1}^{\infty} |A_i|$$

$$\Rightarrow \mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

$\Rightarrow$

μ & δ_x are measures on $(\Omega, \mathcal{F})$.

Q.4. Solve: $\mu_n : \mathcal{F} \to [0, \infty]$ given measure space for $n = 1, 2, \ldots$

$\Rightarrow$ $\mu_n(A) \geq 0 \quad \forall \; A \in \mathcal{F}, \; n \in \mathbb{N}$

Since $\alpha_n \geq 0 \quad \forall \; n \in \mathbb{N}$ given

$\Rightarrow$

$$\alpha_n \cdot \mu_n(A) \geq 0 \qquad \forall \; n \in \mathbb{N}, \; A \in \mathcal{F}$$

$\Rightarrow$

$$\sum_{n=1}^{\infty} \alpha_n \cdot \mu_n(A) \geq 0 \qquad \forall \; n \in \mathbb{N}, \; A \in \mathcal{F}.$$

Now,

Consider $A_1, A_2, \ldots A_n, \ldots \in \mathcal{F}$ be disjoint sets.

Now:

$$\sum_{n=1}^{\infty} \alpha_n \mu_n\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{n=1}^{\infty} \alpha_n \left(\sum_{i=1}^{\infty} \mu_n(A_i)\right)$$

(∵ each μ_n is measure.)

$$= \sum_{n=1}^{\infty} \sum_{i=1}^{\infty} \alpha_n \cdot \mu_n(A_i)$$

$$= \sum_{n=1}^{\infty} \left(\sum_{i=1}^{\infty} \alpha_n \mu_n(A_i)\right)$$

$\Rightarrow$ $\nu := \sum_{n=1}^{\infty} \alpha_n \mu_n$ is a measure function.

Q.5. Ω - finite or countably infinite.

$\Rightarrow$ ~~$A \in$~~ $A \subseteq \Omega$,

Then $A = \bigcup_{\omega_i \in A} \{\omega_i\}$

Now, define $P : 2^{\Omega} \longrightarrow [0, \infty]$

s.t.

$$P(2^{\Omega}) = 1.$$

and

$$P(A) = P\left(\bigcup_{\omega_i \in A} \{\omega_i\}\right)$$

$$= \sum_{\omega_i \in A} P(\omega_i) \qquad \left(\because \bigcup_{\omega_i \in A} \{\omega_i\} \text{ is disjoint union of subsets of } \Omega\right)$$

$$= \sum_{\omega_i \in \Omega} P(\{\omega_i\})\, \delta_{\omega_i}(A)$$

$\Rightarrow$

$$P := \sum_{\omega \in \Omega} P_\omega \delta_\omega \quad ; \text{ where } P_\omega := P(\{\omega\}), \ \forall\, \omega \in \Omega.$$

□□

Q.6. Given (Ω, F, μ) is a measure space. $\{A_n\}_n \in F$ be a sequence of sets in F.

Ⓐ $B_1 = A_1 \Rightarrow B_1 \subseteq A_1$

$B_2 = A_2 \setminus A_1 \Rightarrow B_2 \subseteq A_2$

$B_3 = A_3 \setminus (A_1 \cup A_2) \Rightarrow B_3 \subseteq A_3$

$\vdots$

$B_n = A_n \setminus (A_1 \cup A_2 \cup \dots A_{n-1}) \Rightarrow B_n \subseteq A_n.$

Now:

but $x \in \bigcup_{i=1}^{n} B_i \Rightarrow x \in B_j$ for some j.

since $B_j \subseteq A_j$

$\Rightarrow x \in A_j$

$\Rightarrow x \in \bigcup_{i=1}^{n} A_i$

~~Note that~~

$$\Rightarrow \bigcup_{i=1}^{n} B_i \subseteq \bigcup_{i=1}^{n} A_i \quad \text{---------- (1)}$$

Now:

let $x \in \bigcup_{i=1}^{n} A_i$

$\Rightarrow x \in A_j$ for ~~$j = 1, 2, 3$~~

$j = i_1, i_2, \ldots i_n$ where $i_l \in \mathbb{N}$.

& $i_1 < i_2 < i_3 < \ldots < i_n$

$\Rightarrow x \in A_{i_1} / (A_1 \cup A_2 \cup A_{i_1 - 1})$

$\Rightarrow x \in B_{i_1} \Rightarrow x \in \bigcup_{i=1}^{n} B_i$

$$\Rightarrow \bigcup_{i=1}^{n} A_i \subseteq \bigcup_{i=1}^{n} B_i \quad \text{---------- (2)}$$

from (1) & (2) $\left(\bigcup_{i=1}^{n} A_i = \bigcup_{i=1}^{n} B_i \right)$

for $n \to \infty$

$$\bigcup_{i=1}^{\infty} A_i = \bigcup_{i=1}^{\infty} B_i.$$

iii) Now, by construction, $B_1, B_2, \ldots B_n, \ldots$ are disjoint ~~sets~~ set. So by countable additivity of μ, we have

$$\mu\left(\bigcup_{i=1}^{\infty} B_i \right) = \sum_{i=1}^{\infty} \mu(B_i)$$

Since $B_i \subseteq A_i \Rightarrow \mu(B_i) \leq \mu(A_i)$

$\Rightarrow$ So $\sum_{i=1}^{\infty} \mu(B_i) \leq \sum_{i=1}^{\infty} \mu(A_i)$

$\Rightarrow \mu\left(\bigcup_{i=1}^{\infty} B_i \right) \leq \sum_{i=1}^{\infty} \mu(A_i)$ □□.

Q.7. $(\Omega, \mathcal{F}, P)$ is a probability space.

Since
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Since
$$A \cup B \subseteq \Omega$$
$$\Rightarrow P(A \cup B) \leq P(\Omega)$$
$$= 1$$

So $P(A) + P(B) - P(A \cap B) \leq 1$

$\Rightarrow P(A) + P(B) - 1 \leq P(A \cap B)$

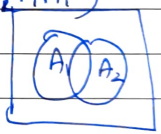
Since $P(A \cap B) \geq 0$

So $P(A \cap B) \geq \max\{0, P(A) + P(B) - 1\}$.

Q.8. $(\Omega, \mathcal{F}, P)$ be given a probability space. We will use the Principal of mathematical induction to prove this.

Let A_1, A_2 be two sets.

$$A_1 \cup A_2 = (A_1 \cap A_2^c) \cup (A_1 \cap A_2) \cup (A_2 \cap A_1^c)$$



$$\Rightarrow P(A_1 \cup A_2) = P(A_1 \cap A_2^c) + P(A_1 \cap A_2) + P(A_2 \cap A_1^c) \quad \text{---(1)}$$

$A_1 = A_1 \cap (A_1 \cap A_2^c)$

$\Rightarrow P(A_1) \leq P(A_1) + 1$

$A_1 = (A_1 \cap A_2^c) \cup (A_1 \cap A_2)$ & $A_2 = (A_2 \cap A_1^c) \cup (A_1 \cap A_2)$

$\Rightarrow P(A_1) - P(A_1 \cap A_2) = P(A_1 \cap A_2^c)$

&

$$P(A_2) - P(A_1 \cap A_2) = P(A_2 \cap A_1^c)$$

put in (1)

$$\Rightarrow P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2).$$

i.e. for $n=1,2,$ statement is true. Let us consider, statement true for $n=k$. i.e.

$$P\left(\bigcup_{i=1}^{k} A_k\right) = \sum P(A_k) - \sum_{i\neq j} P(A_i \cap A_j) \quad (2)$$

$$+ \ldots + (-1)^{k-1} P(A_1 \cap A_2 \ldots \cap A_k)$$

We will prove for $n=k+1$.

$$\Rightarrow P\left(\bigcup_{i=1}^{k+1} A_k\right) = P\left(\left\{\bigcup_{i=1}^{k} A_k\right\} \cup A_{k+1}\right)$$

$$= P\left(\bigcup_{i=1}^{k} A_i\right) + P(A_{k+1}) - P\left(\left\{\bigcup_{i=1}^{k} A_k\right\} \cap A_{k+1}\right)$$

$$= \sum_{k=1}^{k+1} P(A_i) - \sum_{i\neq j}^{k} P(A_i \cap A_j) + \ldots + (-1)^{k+1} P\left(\bigcap_{i=1}^{k} A_i\right)$$

$$- P\left((A_1 \cup A_2 \cup \ldots A_k) \cap A_{k+1}\right) \quad (*)$$

Now:

$$P\left((A_1 \cup A_2 \ldots \cup A_k) \cap A_{k+1}\right) = P\left(\bigcup_{i=1}^{k} (A_i \cap A_{k+1})\right)$$

$$= \sum_{i=1}^{k} P(A_i \cap A_{k+1}) - \sum_{i\neq j}^{k} P(A_i \cap A_j \cap A_{k+1})$$

$$+ \ldots + (-1)^{k-1} P(A_1 \cap A_2 \cap \ldots A_k \cap A_{k+1})$$

apply statement (2) - so put in (*) we get

__/__/____

we get

$$P\left(\bigcup_{i=1}^{k+1} A_k\right) = \sum_{i=1}^{k+1} P(A_i) - \sum_{i\neq j}^{k+1} P(A_i \cap A_j)$$

$$+ \sum_{i\neq j\neq k}^{k+1} P(A_i \cap A_j \cap A_k)$$

$$+ \cdots + \cdots - (-1)^{k-1} P\left(\bigcap_{i=1}^{k+1} A_i\right)$$

$$\Rightarrow P\left(\bigcup_{i=1}^{k+1} A_k\right) = \sum_{i=1}^{k+1} P(A_i) - \sum_{i\neq j}^{k+1} P(A_i \cap A_j)$$

$$+ \cdots + (-1)^k P\left(\bigcap_{i=1}^{k+1} A_i\right).$$

□□